

## LAB EXPERIMENT RECORD

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <b>Department</b>                | AIDS                                      |
| <b>Course Code / Course Name</b> | DSA0601 / Data Handling and Visualization |
| <b>Experiment</b>                | Data Visualization using R                |
| <b>Name</b>                      | DEEKSHITHA U                              |
| <b>Register Number</b>           | 192424055                                 |
| <b>Date</b>                      | 01 October 2026                           |
| <b>Day</b>                       | Tuesday                                   |

## SET 6: PRODUCT SALES ANALYSIS

### Dataset: Monthly Product Sales

The given data represents sales for the first quarter — **January, February, and March**.

| Product ID | Product Name | January Sales | February Sales | March Sales |
|------------|--------------|---------------|----------------|-------------|
| 1          | Product A    | 2000          | 2200           | 2400        |
| 2          | Product B    | 1500          | 1800           | 1600        |
| 3          | Product C    | 1200          | 1400           | 1100        |

### Aim

To analyze monthly product sales for the first quarter using grouped bar chart, stacked area chart, and a table, and to develop an interactive Tableau dashboard for exploring the sales data.

## PROGRAM 1: Grouped Bar Chart of Product Sales

### Question

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

### R Program

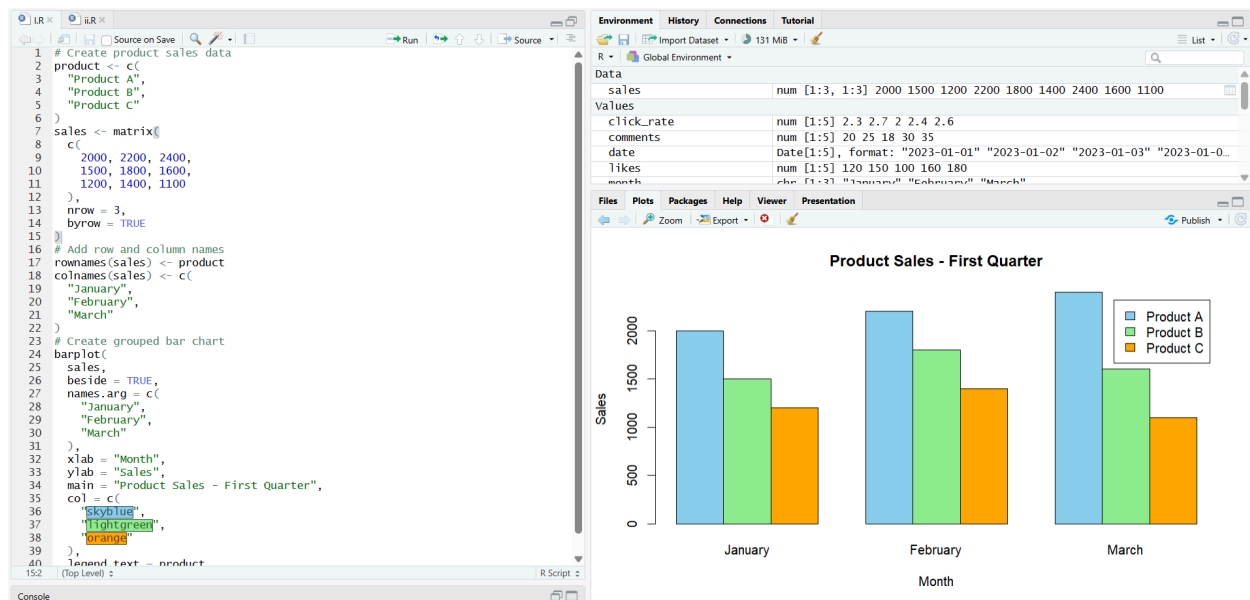
```
# =====  
# PROGRAM 1: GROUPED BAR CHART  
# =====  
  
# Create product sales data  
product <- c(  
  "Product A",  
  "Product B",  
  "Product C"  
)  
  
sales <- matrix(  
  c(  
    2000, 2200, 2400,  
    1500, 1800, 1600,  
    1200, 1400, 1100  
  ),  
  nrow = 3,  
  byrow = TRUE  
)  
  
# Add row and column names  
rownames(sales) <- product  
  
colnames(sales) <- c(  
  "January",  
  "February",  
  "March"  
)  
  
# Create grouped bar chart  
barplot(  
  sales,
```

```

    beside = TRUE,
    names.arg = c(
      "January",
      "February",
      "March"
    ),
    xlab = "Month",
    ylab = "Sales",
    main = "Product Sales - First Quarter",
    col = c(
      "skyblue",
      "lightgreen",
      "orange"
    ),
    legend.text = product
  )

```

## Output:



## Observation

Product A has the highest sales among the three products in all three months. Product C has the lowest sales in January and February and also has the lowest sales in March. Product A shows an increasing sales trend from January to March.

## PROGRAM 2: Stacked Area Chart of Overall Sales Trend

### Question

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

### R Program

```
# =====  
# PROGRAM 2: STACKED AREA CHART  
# =====  
  
# Months  
month <- c(  
  "January",  
  "February",  
  "March"  
)  
  
# Product sales  
product_A <- c(  
  2000,  
  2200,  
  2400  
)  
  
product_B <- c(  
  1500,  
  1800,  
  1600  
)  
  
product_C <- c(  
  1200,  
  1400,  
  1100  
)  
  
# Calculate cumulative sales  
total_A <- product_A
```

```
total_AB <- product_A + product_B
```

```
total_ABC <- product_A + product_B + product_C
```

```
# Create empty plot
```

```
plot(  
  1:3,  
  total_ABC,  
  type = "n",  
  xaxt = "n",  
  xlab = "Month",  
  ylab = "Sales",  
  main = "Overall Product Sales Trend",  
  ylim = c(  
    0,  
    max(total_ABC)  
  )  
)
```

```
# Add month labels
```

```
axis(  
  1,  
  at = 1:3,  
  labels = month  
)
```

```
# Product A area
```

```
polygon(  
  c(1:3, 3:1),  
  c(  
    total_A,  
    rep(0, 3)  
  ),  
  col = "skyblue",  
  border = "blue"  
)
```

```
# Product B area
```

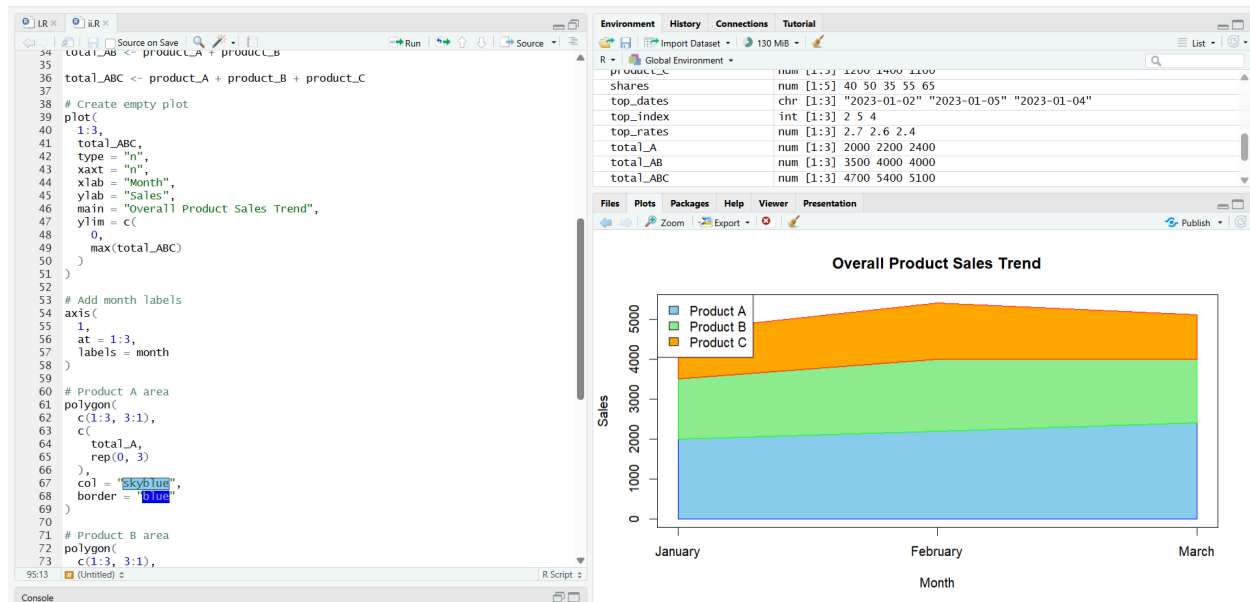
```
polygon(
```

```
    c(1:3, 3:1),  
    c(  
      total_AB,  
      rev(total_A)  
    ),  
    col = "lightgreen",  
    border = "green"  
  )  
)
```

```
# Product C area  
polygon(  
  c(1:3, 3:1),  
  c(  
    total_ABC,  
    rev(total_AB)  
  ),  
  col = "orange",  
  border = "red"  
)
```

```
# Add legend  
legend(  
  "topleft",  
  legend = c(  
    "Product A",  
    "Product B",  
    "Product C"  
  ),  
  fill = c(  
    "skyblue",  
    "lightgreen",  
    "orange"  
  )  
)
```

# Output



## Observation

The total sales were 4700 in January, 5400 in February, and 5100 in March. Overall sales increase from January to February and decrease slightly in March. Product A contributes the largest share of total sales during all three months.

## PROGRAM 3: Table of Monthly Product Sales

### Question

Build a table to show the monthly sales data for each product. Label the table elements.

### R Program

```
# =====  
# PROGRAM 3: MONTHLY SALES TABLE  
# =====  
  
# Create monthly sales table  
sales_table <- data.frame(  
  Product_ID = c(  
    1,  
    2,  
    3  
  ),  
  
  Product_Name = c(  
    "Product A",  
    "Product B",  
    "Product C"  
  ),  
  
  January_Sales = c(  
    2000,  
    1500,  
    1200  
  ),  
  
  February_Sales = c(  
    2200,  
    1800,  
    1400  
  ),  
  
  March_Sales = c(  
    2400,  
    1600,
```

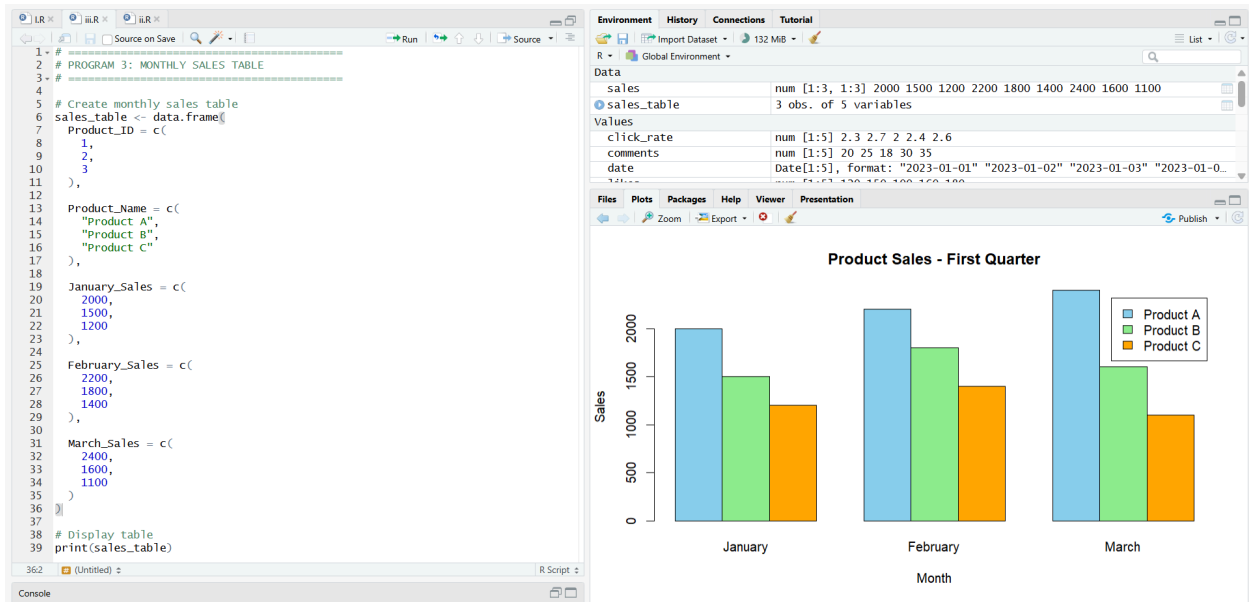
```

1100
)
)

# Display table
print(sales_table)

```

## Output



## Observation

The table presents the monthly sales values for each product during the first quarter. Product A records the highest sales in all three months.

## PROGRAM 4: Tableau Dashboard

### Question

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

#### Step 1 — Prepare Excel Data

Create an Excel file with this data:

| Product ID | Product Name | January Sales | February Sales | March Sales |
|------------|--------------|---------------|----------------|-------------|
| 1          | Product A    | 2000          | 2200           | 2400        |
| 2          | Product B    | 1500          | 1800           | 1600        |
| 3          | Product C    | 1200          | 1400           | 1100        |

Save it as:

*Product\_Sales.xlsx*

#### Step 2 — Open Tableau

Open **Tableau Desktop** or **Tableau Public**.

Select:

**Connect → Microsoft Excel**

Then select:

*Product\_Sales.xlsx*

#### Step 3 — Create the Grouped Bar Chart

Create **Sheet 1**.

For Tableau, the easiest way to work with January, February, and March is to **pivot** the three sales columns.

In the Data Source page:

Select:

January Sales

February Sales

March Sales

Right-click → **Pivot**.

Rename:

Pivot Field Names → Month

Pivot Field Values → Sales

You will now have:

| <b>Product Name</b> | <b>Month</b>   | <b>Sales</b> |
|---------------------|----------------|--------------|
| Product A           | January Sales  | 2000         |
| Product A           | February Sales | 2200         |
| Product A           | March Sales    | 2400         |
| Product B           | January Sales  | 1500         |
| ...                 | ...            | ...          |

Then create the chart:

Drag:

Month



Columns

Drag:

Sales



Rows

Drag:

Product Name



Color

Select:

**Marks → Bar**

This gives you a grouped bar chart comparing the three products across the three months.

Title:

***Product Sales – First Quarter***

#### **Step 4 — Create the Stacked Area Chart**

Create **Sheet 2**.

Drag:

Month



Columns

Drag:

Sales



Rows

Change:

**Marks → Area**

Then drag:

Product Name



Color

This creates a **stacked area chart** showing the contribution of each product to total sales.

Title:

***Overall Product Sales Trend***

### **Step 5 — Create the Sales Table**

Create **Sheet 3**.

Drag:

Product Name



Rows

Then drag:

Month



Columns

Drag:

Sales



Text

This creates a table showing the sales of each product for each month.

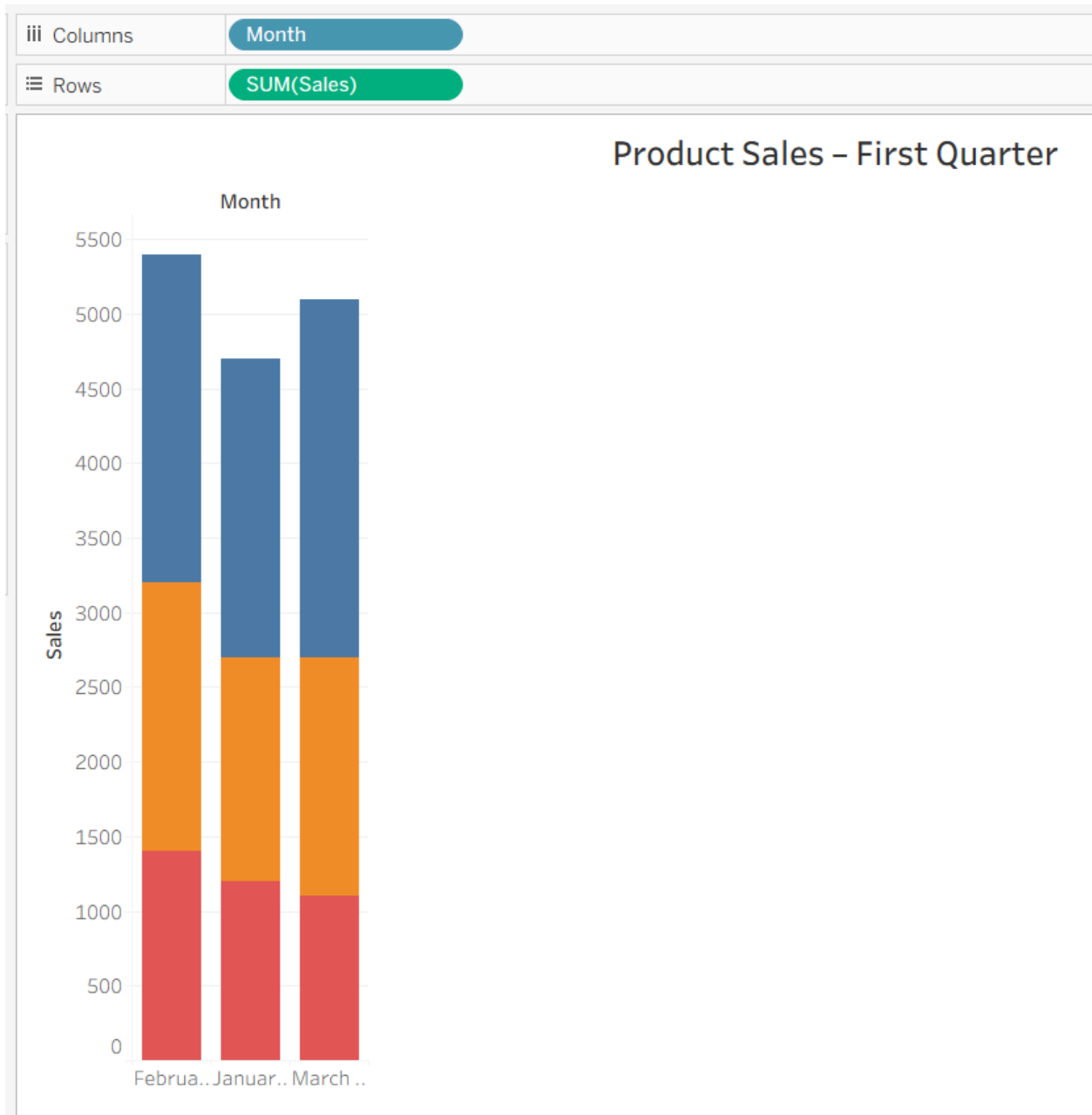
Title:

***Monthly Product Sales***

## Step 6 — Create the Tableau Dashboard

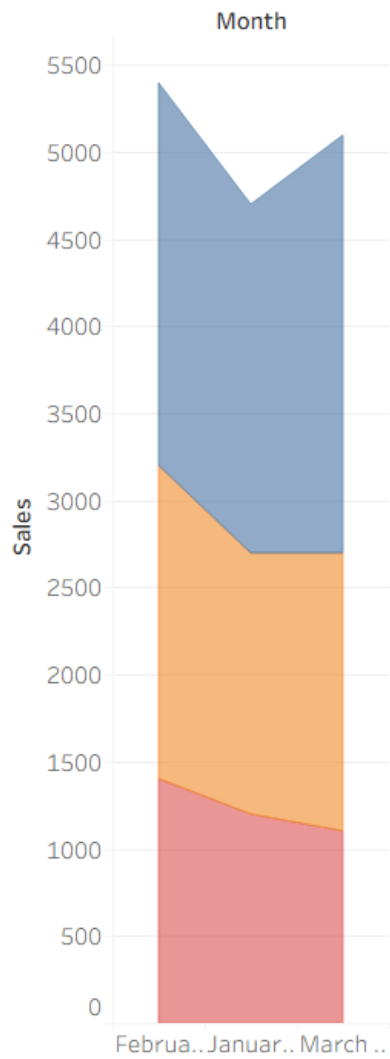
Click:

### New Dashboard

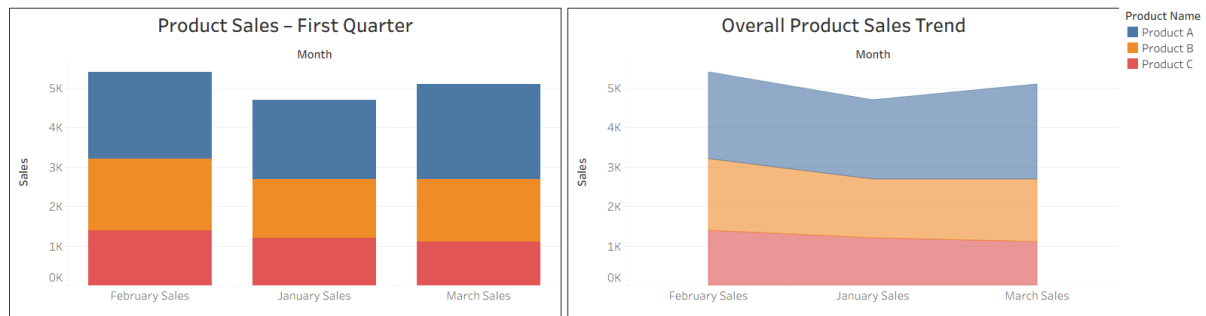


|         |            |
|---------|------------|
| Columns | Month      |
| Rows    | SUM(Sales) |

## Overall Product Sales Trend



|                       |                        |
|-----------------------|------------------------|
| Columns               | Month                  |
| Rows                  | Product Name           |
| Monthly Product Sales |                        |
| Month                 |                        |
| Product Name          | Febru.. Janua.. Marc.. |
| Product A             | 2,200 2,000 2,400      |
| Product B             | 1,800 1,500 1,600      |
| Product C             | 1,400 1,200 1,100      |



|                       |                |               |             |
|-----------------------|----------------|---------------|-------------|
| Monthly Product Sales |                |               |             |
| Month                 |                |               |             |
| Product Name          | February Sales | January Sales | March Sales |
| Product A             | 2,200          | 2,000         | 2,400       |
| Product B             | 1,800          | 1,500         | 1,600       |
| Product C             | 1,400          | 1,200         | 1,100       |

# Observation

The Tableau dashboard combines the grouped bar chart, stacked area chart, and monthly sales table. Users can interact with the dashboard using product and month filters to explore sales performance.

## Procedure

1. Create the product sales data for January, February, and March.
2. Store the product sales values in R.
3. Use `barplot()` with `beside = TRUE` to create the grouped bar chart.
4. Calculate cumulative sales values for the stacked area chart.
5. Use `polygon()` to create the stacked areas.
6. Create a data frame containing the monthly sales of each product.
7. Use `print()` to display the sales table.
8. Prepare the product sales data in Excel.
9. Import the Excel file into Tableau.
10. Pivot the January, February, and March sales columns.
11. Create the grouped bar chart in Tableau.
12. Create the stacked area chart in Tableau.
13. Create the monthly sales table.
14. Combine all three worksheets into a Tableau dashboard.
15. Add filters for interactive exploration.

## Result

The first-quarter product sales data was successfully visualized using a grouped bar chart and stacked area chart. A table was created to display the monthly sales of each product. An interactive Tableau dashboard was also developed by combining the grouped bar chart, stacked area chart, and sales table.